

```
meta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953
total 2000
ballocc: first 1101 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -
nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=
raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sandbox 32768 - cat README
cat: cannot open README
$ sandbox 2 - sbtest fork
sbtest: fork ret -1
sbtest: alive
$ sandbox 32768 README grep xv6 README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix
Version 6 (v6).  xv6 loosely follows the structure and style of v6,
xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer
(kaashoek,rtm@mit.edu).  The main purpose of xv6 is as a teaching
$ sandbox 32768 README grep xv6 x
grep: cannot open x
$ sandbox 32768 - sbtest escape
sbtest: escape confined
sbtest: alive
$ sandbox 32768 README sbtest forkpath
sbtest: forkpath child README ok
sbtest: forkpath child exec.sh denied
sbtest: alive
$ sandbox 2048 - sbtest getpid
sbtest: getpid ret -1
sbtest: alive
$ echo SandBox_Q1_Q2
SandBox_Q1_Q2
$ echo Sudheer_Rathore_31683
Sudheer_Rathore_31683
$ █
```

```
sudh@sudh-HP-Laptop-15s-fq5xxx:~/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep
user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stress
fs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan
user/_dorphan user/_sync user/_attack user/_secret user/_sandbox user/_sbtest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953
total 2000
balloc: first 1101 blocks have been allocated
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raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ secret xyzzy
$ attack
xyzzy
$ echo Attack_Q3
Attack_Q3
$ echo Sudheer_Rathore_31683
Sudheer_Rathore_31683
$ █
```

```
S
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/sudh/xv6-labs-2026'
== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (2.7s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (0.2s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (1.1s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (0.9s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.0s)
== Test sandbox_block ==
$ make qemu-gdb
sandbox_block: OK (1.0s)
== Test sandbox_escape ==
$ make qemu-gdb
sandbox_escape: OK (1.0s)
== Test sandbox_forkpath ==
$ make qemu-gdb
sandbox_forkpath: OK (1.0s)
== Test attack ==
$ make qemu-gdb
attack: OK (1.1s)
== Test time ==
time: OK
Score: 60/60
sudh@sudh-HP-Laptop-15s-fq5xxx:~/xv6-labs-2026$ echo Sudheer_Rathore_31683
Sudheer_Rathore_31683
sudh@sudh-HP-Laptop-15s-fq5xxx:~/xv6-labs-2026$
```